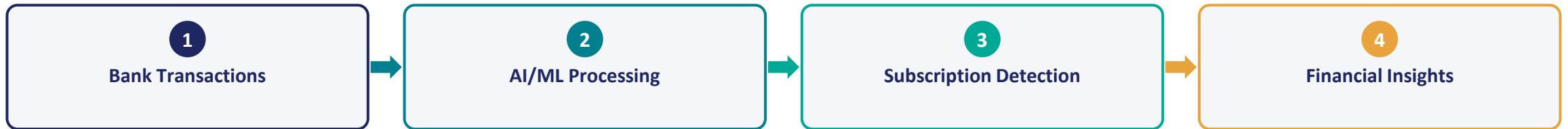


SUBSCRIPTION LEAK DETECTOR

An AI/ML-Powered Recurring Subscription Detection
and Financial Leak Analysis System



PROJECT TEAM

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Academic Session 2023–2027

Problem Statement

Consumers increasingly rely on:

- Streaming platforms
- SaaS products
- Cloud storage
- Fitness applications
- Other recurring services

Bank descriptions are often:

Short

Inconsistent

Cryptic

Difficult to interpret

"NETFLIX.COM 8829 CA"

vs.

"NETFLIX*STREAM"

Existing budgeting apps show spending but do not reliably identify:

- ✗ Recurring charges
- ✗ Merchant-name variations
- ✗ Silent price increases
- ✗ Subscription review candidates

Messy Transactions

Hidden Recurring Payments

Difficult Manual Tracking

Proposed Solution

How can we automatically identify recurring subscription-like payments from noisy bank transaction data and provide transparent, actionable insights?

Literature Review

Research Direction	Approach	Limitation
Subscription / Recurring Transaction Detection	Identify recurring transactions using frequency, periodicity, and payment history.	Merchant descriptions are noisy and inconsistent.
Merchant Entity Resolution	Text normalization and similarity matching identify the same merchant across variant descriptions.	Exact matching fails with transaction IDs, abbreviations, extra text.
Explainable AI / Transparent Scoring	Exposes the factors contributing to a prediction or score.	Users need to understand why a transaction is flagged.
Price-Creep / Price-Increase Analysis	Historical amounts reveal changes in recurring payments over billing cycles.	Users often overlook gradual price increases.
User Feedback / Adaptive Systems	User feedback improves future decisions or scoring.	Confirmed/cancelled feedback can refine subsequent scoring.

RESEARCH GAP

Existing approaches often address recurring transaction detection, merchant matching, or explainability separately. This project combines merchant entity resolution, statistical recurrence detection, explainable confidence scoring, price-creep detection, and user feedback into one end-to-end system.

Project Objectives

01

Automate recurring
subscription detection

02

Resolve messy
merchant names

03

Track silent
price increases

04

Explainable Subscription
Confidence Score

05

Interactive savings
simulator

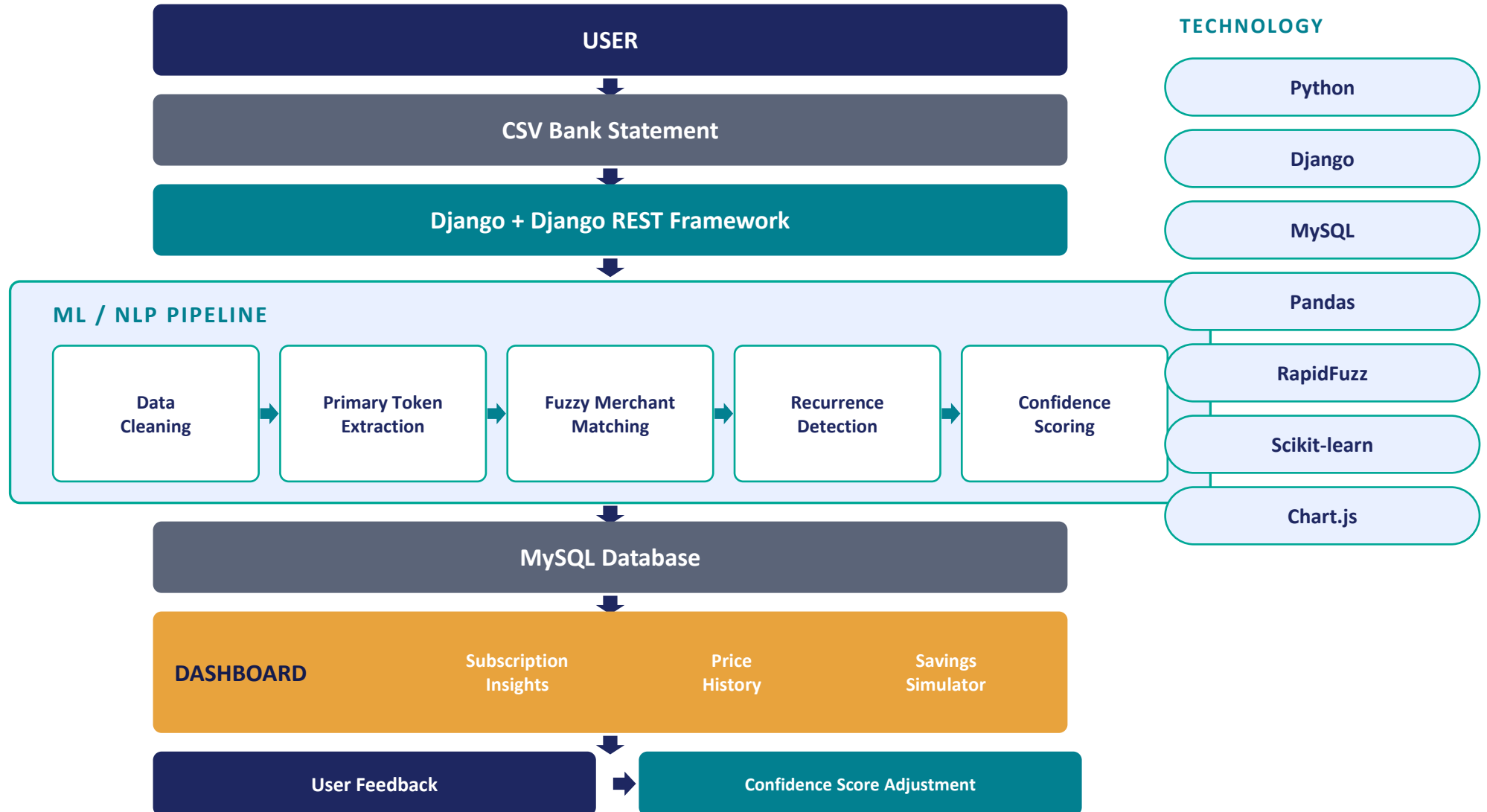
06

Collect and use
user feedback

07

Clear, actionable
dashboard

Proposed Solution & System Architecture



ML / NLP Pipeline



Merchant Entity Resolution & Recurring Detection

Merchant Matching

NETFLIX.COM 8829

NETFLIX*STREAM

NETFLIX.COM CA

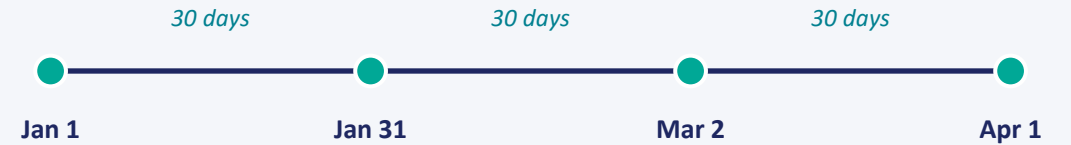
NFLX STREAMING



NETFLIX

- Exact matching treats variants as different merchants.
- Fuzzy matching groups similar descriptions together.
- Final approach: `partial_ratio` applied to the primary brand token.

Recurrence Detection



Transaction Count ≥ 3

+

Gap Std. Dev. ≤ 5 days

+

Mean Gap $\approx 7 / 30 / 365$ days



Recurring Candidate

Feature Engineering & Subscription Confidence Score

Feature Vector

Transaction Count	Gap Mean
Gap Std. Deviation	Amount Mean
Amount Std. Deviation	Price Increase %
Days Since Last Charge	

SUBSCRIPTION CONFIDENCE SCORE

$$C = \min(1.0, b + w_1I_1 + w_2I_2 + w_3I_3)$$

The score measures how strongly a recurring payment pattern resembles a subscription worth reviewing.

Weight Components



Note: This is called the **Subscription Confidence Score** — never a “Risk Score” — because it reflects payment-pattern confidence, not financial danger.

Explainable Insights, Price-Creep Detection & Feedback

1 · EXPLAINABILITY

Subscription
Confidence Score

0.80

WHY?

- ✓ Recurring payment pattern confirmed
- ✓ Price increased by 27%
- ✓ Payment amount is relatively stable

2 · PRICE CREEP

Cycle 1–6 ₹199

Cycle 7–13 ₹249 +25.1%

Cycle 14–18 ₹299 +20.1%

Increase Events > 2%

+

Cumulative Growth > 10%

+

No Single Increase > 50%



Price Creep Detected

3 · USER FEEDBACK

Detected Subscription



User Review



Confirmed / Cancelled



Future Score Adjustment

Cancelled	+0.15
Confirmed / Still in Use	-0.20
Mixed / No Feedback	0

This is feedback-informed heuristic adjustment, not model retraining.

Technology Stack & Database

F**Frontend**

HTML5 • CSS3 • JavaScript • Chart.js / Recharts

B**Backend**

Python • Django • Django REST Framework

M**ML / NLP**

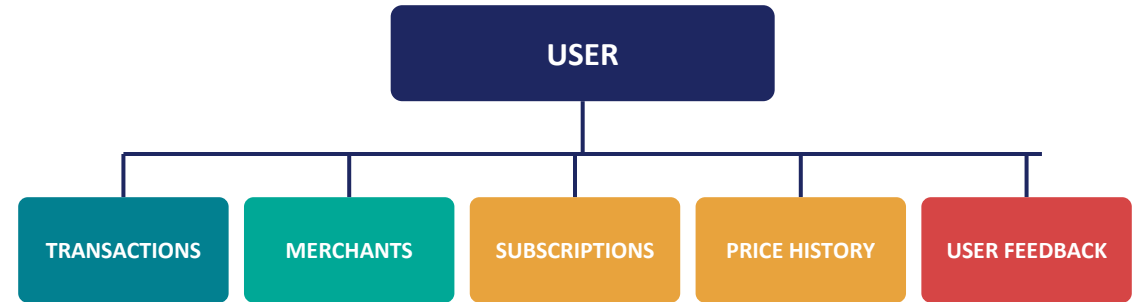
Pandas • RapidFuzz • Scikit-learn

D**Database**

MySQL • MySQL Workbench

D**Development / Data**

Faker • Jupyter Notebook • VS Code • Git • GitHub



Transactions → Merchants → Subscriptions → Price History

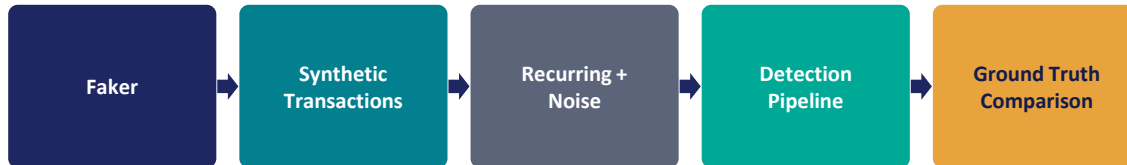
Subscriptions → User Feedback

Relational schema keeps every transaction traceable from raw bank line item through to a scored, explainable subscription record — scoped per logged-in user.

Dataset, Evaluation & Results

DATASET

- Synthetic bank transaction data generated using Faker
- 6 predefined recurring merchants
- Weekly, monthly, and annual billing frequencies
- Scheduled price increases
- ~80 random one-off transactions as noise
- Ground-truth labels kept separate from the pipeline



ADDITIONAL FINDINGS

- ✓ Zero false positives observed in the test set
- ✓ 5 of 6 known recurring merchants were detected
- ✓ One annual subscription missed — only two transactions existed within the observation window

RESULTS

100%

Precision

~83%

Recall

~0.90

F1 Score

100

83

90

Precision

Recall

F1 Score

Limitations, Future Scope & Conclusion

LIMITATIONS

- ✗ No real bank API integration currently
- ✗ Transaction data cannot reveal actual service usage
- ✗ Annual subscriptions harder to detect with limited observations
- ✗ Rent, EMI, utilities, etc. can resemble subscriptions
- ✗ Current confidence weights are heuristic

FUTURE SCOPE

- ✓ Real bank API integration
- ✓ Better merchant embeddings
- ✓ Logistic Regression / Random Forest comparison
- ✓ Periodic supervised model retraining
- ✓ Mobile application
- ✓ Advanced price benchmarking
- ✓ Improved low-frequency subscription detection

CONCLUSION

Subscription Leak Detector transforms noisy bank transaction data into transparent and actionable subscription insights through merchant entity resolution, statistical recurrence detection, explainable confidence scoring, price-creep detection, and savings simulation.



THANK YOU

Questions & Answers

Subscription Leak Detector

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Bank Statement



AI/ML Pipeline



Subscription Insights